

CAPACITIVE STRUCTURAL HEALTH MONITORING OF HYBRID METAL–COMPOSITE JOINTS

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Abstract

Hybrid metal-composite joints are increasingly used in lightweight structures for automotive and marine applications, yet their reliability is often compromised by interfacial damage such as delamination or adhesive failure. This study presents a multi-channel Structural Health Monitoring (SHM) approach for hybrid metal-GFRP joints based on capacitive self-sensing. The sensing principle relies on a capacitor-like configuration embedded in the joint: the metal adherend acts as one electrode, while thin metallic foils applied or co-cured on the composite surface form the second. An array of these foils enables localized monitoring by detecting capacitance variations driven by dielectric and geometric changes during damage propagation. A distinctive feature of this work is the synergistic interaction between the sensing system and additively manufactured lattice structures at the interface. These 3D architectures, originally designed to enhance mechanical interlocking and joint strength, also amplify the sensitivity to interfacial separation. During Mode I fracture testing (Double Cantilever Beam), the advancement of the crack front triggers the sequential fracture and instantaneous electrical disconnection of the metallic lattice struts. This transition from a grounded to a floating state produces discrete, localized capacitance drops, enabling real-time spatial tracking of the crack tip. The experimental campaign validates this approach through preliminary dielectric characterization and synchronized mechanical-capacitive testing, utilizing a custom high-resolution acquisition system. The results demonstrate that this capacitive technology provides a promising, non-intrusive solution for monitoring damage evolution in high-performance hybrid joints, reducing the need for periodic inspections.

Keywords: *Hybrid metal-composite joints; Lattice reinforcements; Structural health monitoring; Capacitive sensing; Additive manufacturing.*

1 Introduction

The evolution of engineering design in high-performance sectors (aerospace, automotive, and marine) is currently dominated by the lightweight design paradigm. The need to reduce emissions has driven research toward multi-material architectures, combining high-strength

metal alloys with fiber-reinforced polymers (FRP). However, the coexistence of materials with divergent properties makes designing reliable metal-composite hybrid joints a critical challenge. Direct assembly and polymer-metal hybrid manufacturing routes have therefore been increasingly investigated [1]. Traditional joining technologies fail to fully exploit the potential of hybrid systems. Purely mechanical joints (bolting, riveting) interrupt fiber continuity, generating severe stress concentrations and machining-induced delamination. Conversely, structural adhesive bonding distributes stress but is heavily dependent on surface chemistry, vulnerable to hygrothermal aging, and prone to sudden, catastrophic failure without prior plastic deformation [2]. To overcome these limitations, through-the-thickness reinforcement strategies have been developed [3]. Early penetrative concepts, such as HYPER pins, demonstrated that metallic features protruding from the adherend can delay damage initiation, promote progressive failure, significantly increasing joint strength [4]. Subsequent studies on additively manufactured pins linked the macroscopic fracture response to the pull-out and local support conditions of individual reinforcements [5, 6]. Advanced Additive Manufacturing (AM), specifically Laser Powder Bed Fusion (LPBF), integrates complex metallic interfaces (3D lattices, pyramids, pin arrays) directly onto metal substrates for subsequent composite infiltration via co-curing, compression, injection, or overmolding. Deep Interface Co-Molding of SMC within pyramidal SLM lattices showed significantly increased pull-out load and absorbed energy versus flat interfaces [7]. Recent studies have confirmed that 3D printed pins, AlSi10Mg surface structures and micro-penetrative anchors can improve interfacial toughness, lap-shear strength, fatigue performance and robustness [8, 9, 10]. This interlocking significantly increases contact area, shifting 2D linear crack propagation into a tortuous, dissipative 3D trajectory involving local bridging, strut fracture, and composite delamination. However, stiff metallic features create extreme stiffness gradients, and thermal expansion mismatches generate post-production residual stresses, making Mode I fracture test interpretation highly complex. In this context, Structural Health Monitoring (SHM) becomes a primary damage tolerance requirement to prevent hidden damage from evolving into sudden joint failure [11]. While various SHM technologies have been explored, reliable localization of bond-line damage remains challenging [12]. Fiber Bragg gratings (FBG) are mature optical sensors but require complex strain-temperature compensation [13]. Carbon-coated piezoresistive fibers and nanomaterial-functionalized fabrics offer structural monitoring, but they require matrix or reinforcement modifications and careful conductive network control [14]. Capacitive sensing is ideal for this hybrid architecture, as the metallic interlocking features and external electrodes directly act as the sensing system. Previous research on GFRP has shown that capacitance measurements can detect matrix cracking [15], monitor curing quality and localize loading [16], and identify impact-induced delaminations in composite materials using fringing electric fields [17]. Based on this background, the present work investigates a hybrid metal-GFRP joint in which additively manufactured pyramidal lattices provide both mechanical toughening and functional sensitivity to interfacial damage. The metal substrate acts as a grounded primary electrode, the GFRP acts as a dielectric layer, and a multi-pad copper array on the outer composite surface defines localized sensing regions. As the Mode I crack front advances, the mechanical fracture of the pyramidal lattice causes an instantaneous transition of the local metallic features from grounded to floating conductive elements. This physical disconnection produces discrete, step-wise capacitance drops that can track crack propagation. The core objective of this study is to demonstrate the interlocking architecture as a true multifunctional feature, successfully combining 3D mechanical reinforcement with real-time, multi-channel capacitive SHM.

2 Materials and Methods

2.1 Materials

The metallic adherends were manufactured from AlSi10Mg aluminum alloy powder, selected for its high strength-to-weight ratio and suitability for lightweight structural applications. The composite material is a commercial SMC designated as SMC LS 3016 R30 7016, supplied by Polynt S.p.A.. This structural-grade SMC consists of an unsaturated polyester resin matrix reinforced with a 30% mass fraction of glass fibers. Cured declared properties: tensile strength/modulus of 80 MPa/11 GPa, and flexural strength/modulus of 180 MPa/10 GPa.

2.2 Samples Production

The specimens, featuring 3D pyramidal lattice structures, were produced using LPBF technology. Manufacturing was carried out using a Nikon system 180 PS. Square-based pyramidal configurations were produced. Based on previous studies on lattice infiltration and micro-penetrative anchors [7], [10], the following design parameters were adopted to ensure structural integrity: a pyramid size defined by a 6 mm circumscribed base diameter, a nominal strut diameter of 0.8 mm, and an inclination angle of 60° relative to the build platform, resulting in a tip height of 5.88 mm. Two different specimen geometries were produced: a 45x45 mm square geometry featuring 36 pyramids in a 6x6 configuration (x-y spacing of 7 mm) (Figure 1A), and a 25x180 mm rectangular geometry featuring 75 pyramids in a 3x25 configuration (x-y spacing of 7mm) (Figure 1B). The pyramidal structures were printed on a 15 mm thick monolithic base, also printed on AlSi10Mg. The specimens were subsequently cleaned via sand blasting. The compression molding and infiltration process was designed to compress the composite material at elevated temperatures against the metallic component to promote fiber penetration beneath the lattice structures. The metallic components were placed inside custom-made aluminum molds equipped with heating resistors, preheated to 140 °C. The mold for the square specimens featured a cylindrical cavity with a 70 mm diameter, whereas the mold for the rectangular specimens featured a 50x230 mm rectangular cavity. The pre-impregnated composite material was prepared and inserted into the mold over the metallic substrate. For the square specimens, the composite was cut into a 15 x 450 mm strip and wound in a spiral configuration to form a disc matching the mold diameter. For the rectangular specimens, n = 15 parallel strips of 13 x 230 mm were placed inside the mold. The strip dimensions were calculated to achieve a final composite thickness of 12 mm while their positioning was selected in order to facilitate the migration of the fibers below the 3D lattice structures, ensuring a good mechanical interlock. A punch was used to press the composite material against the aluminum substrate at a displacement rate of 5 mm/min until a pressure of 0.25 MPa was reached. The rate was then reduced to 1 mm/min until reaching 6.5 MPa, after which the curing stage was carried out for 30 minutes at 140 °C.

2.3 Characterization of Joint Adhesion and Tensile Separation

Previous studies developed 3D lattice structures to enhance metal-composite joint adhesion, focusing primarily on carbon fiber composites [7], [8], [10]. To validate their effectiveness also with glass fiber composites, preliminary pull-off tests were conducted, comparing (n=3) square-based specimens featuring 3D pyramidal lattices against flat reference specimens. Interfacial tensile strength was evaluated using a servo-hydraulic testing machine (MTS 810) under displacement control at a crosshead speed of 0.2 mm/min. Testing employed a custom fixture consisting of a rigid metal plate to constrain the upper composite component, while a flanged attachment secured the lower aluminum substrate to the pulling tool. For each configuration, the maximum sustained load prior to complete joint failure was recorded.

2.4 Capacitance – Thickness Relationship for Lattice-Integrated dielectric composite

This experiment investigated the influence of the pyramidal array on the dielectric behavior of the system. For this purpose, two square specimens, each featuring a dielectric GFRP layer with an initial thickness of 17 mm, were fabricated: one sample with the 6x6 3D lattice pyramidal array, while the second, without pyramids, served as reference. The thickness of the GFRP substrate was progressively reduced using a CNC milling machine with incremental steps (initially of 1.5 mm, then reduced to 0.5 mm and 0.2 mm while approaching the lattice tips). After every material removal, the equivalent capacitance of the system was characterized using a GW Instek LCR-6002 precision meter (Figure 1C). The measurement was conducted in Cp - Rp (Parallel Capacitance - Parallel Resistance) mode, at $V = 2V$ and $f = 2\text{ kHz}$, using a four-wire Kelvin connection: the "Force" lead pair was connected to the top electrode (circular copper plate, diameter = 45 mm, thickness = 1 mm), while the Sense lead pair was connected to the metallic base of the specimen. This procedure was replicated identically for both specimens to isolate and quantify the capacitive contribution strictly related to the presence of the pyramidal structures.

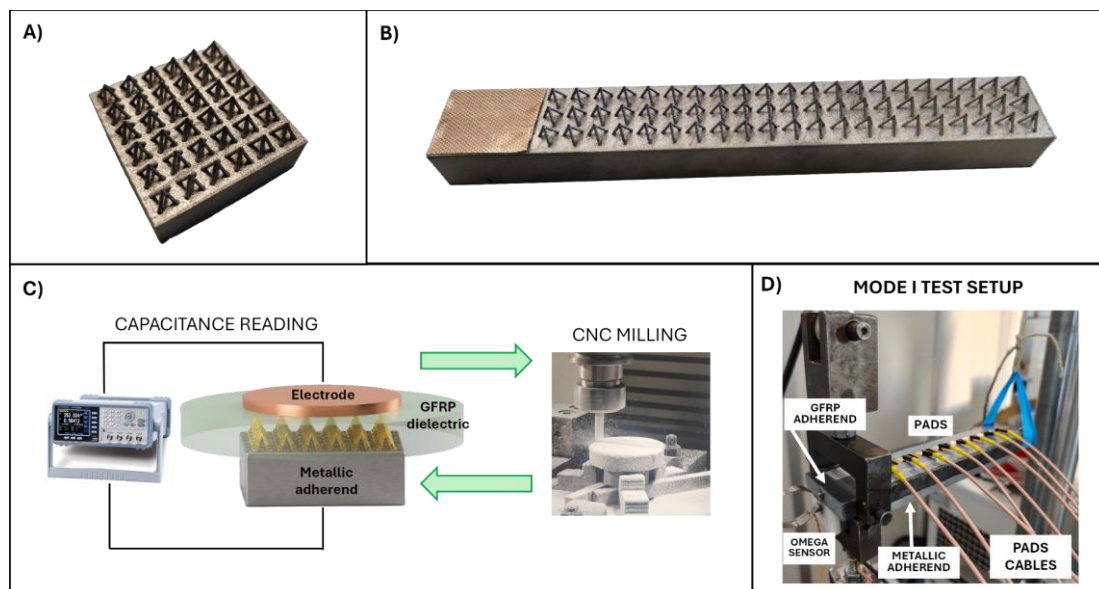


Figure 1. Experimental workflow and setup: A) square and B) rectangular metallic adherend with 3D pyramidal lattice; C) schematics of the capacitive investigation of the samples through incremental CNC milling of the GFRP substrate interspersed by capacitance measurements; D) experimental setup for Mode I DCB testing, showing the multi-pad array and coaxial cabling connected to acquisition system.

2.5 Multi-Channel Capacitive Sensing of Crack Propagation

In order to investigate stable Mode I crack propagation at the hybrid metal-composite interface, a modified Double Cantilever Beam (DCB) configuration was employed, consistent with fracture-mechanics approaches commonly used for hybrid metal-composite joints [5]. A multi-pad capacitive architecture was implemented to achieve high spatial resolution and monitor the crack progression. The sensing principle relies on the progressive failure of the metallic pyramids. As the crack advances, the failure of these individual lattice structures causes their instantaneous electrical disconnection from the grounded adherend (the metallic base), transitioning them to a floating state. This discrete and sudden change produces localized, step-

wise drops in capacitance beneath the above corresponding sensing pad, enabling tracking of the crack front. The custom-made acquisition system is governed by an ESP32 microcontroller, controlling two Texas Instruments FDC1004 capacitance-to-digital converters (4 channels each) that provide a 0.5 fF resolution essential for detecting minute variations during slow crack growth. A digital multiplexer manages I2C communication between the two FDC1004 and the ESP32. Data logging and synchronized polling of the sensing pads are handled via custom-developed software. To neutralize parasitic crosstalk between adjacent pads, hardware-level active shielding was implemented, forcing the electric field into a strictly vertical path through the dielectric. The pads were connected through RG316 coaxial cables, with the metallic sheath driven at the same instantaneous potential of the pad to eliminate horizontal field dispersion. The metallic adherend, with the pyramids, is connected to ground. The multi-pad array was realized using eight 1 mm-thick copper electrodes, each with a width of 15.4 mm and an isolation gap of 3.0 mm between consecutive pads, bonded to the GFRP surface. The capacitance is provided as the variation of each pad from its value at rest (ΔC). A single specimen with rectangular geometry (proof of concept) was selected to ensure a unidirectional, stable Mode I crack propagation, avoiding three-dimensional boundary effects at the crack front. Furthermore, a GFRP thickness of 12 mm was chosen as the optimal compromise between maximizing the capacitive signal sensitivity and maintaining sufficient structural stiffness, as determined by the preliminary tests detailed in Section 2.4. Mode I loading was obtained via parallel 6 mm diameter holes drilled and vertically aligned through both the GFRP and the aluminum substrate to accommodate the pins of the MTS universal testing machine (Figure 1D). To facilitate crack initiation, the first four rows of pyramids (for a total length of 20 mm) were cut prior to the joint production, and the resulting surface was covered with a PTFE adhesive (Figure 1B – left). PAD 1 was positioned directly above the pre-crack tip. An omega-type extensometer was installed between the GFRP and aluminum layers to monitor the crack opening displacement, using a thin plastic spacer as insulation to prevent electrical interference with the aluminum ground (Figure 1D). Displacement-controlled tests were conducted at a constant rate of 2.5 mm/min until total joint separation occurred.

3 Results and Discussion

3.1 Characterization of Joint Adhesion and Tensile Separation

Figure 2A shows the load-displacement curves of the pull-off tests comparing flat interfaces and 3D pyramidal lattice structures.

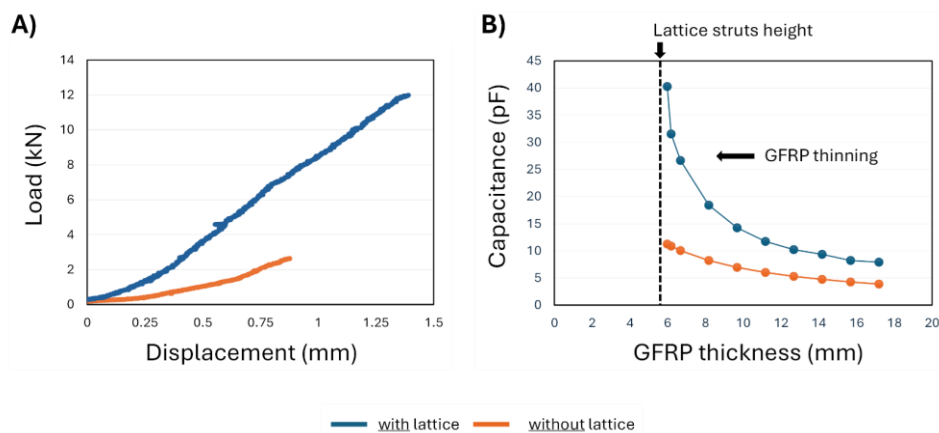


Figure 2. Preliminary characterization of the hybrid joints: A) pull-off tests with and without 3D pyramidal lattice; B) capacitance – thickness relationship following dielectric (GFRP) thinning.

The integration of the 3D pyramidal lattice, consisting of 36 pyramids, led to a significant improvement in mechanical properties, in agreement with previous reports on lattice- and pin-reinforced hybrid interfaces [4], [7], [8]. The average pull-off force increased from 2.1 ± 0.4 kN for the flat specimens to 11.8 ± 2.1 kN for the lattice specimens. This represents a percentage increase of more than 450%, demonstrating the high efficiency of the mechanical interlocking provided by the 3D printed lattice structures. The failure mechanism for the lattice joints was characterized by the detachment of the pyramids at their base. The flat specimens provide a reference value for the nominal adhesion-dominated contribution of the polyester-based SMC to the AlSi10Mg surface, quantified at 1.04 ± 0.24 MPa. In contrast, the lattice configuration reached a total mean stress of 5.8 ± 1.0 MPa. This superior performance is the result of the synergistic action between the resin bonding and the mechanical interlocking of the glass fibers infiltrated beneath the pyramidal lattice.

3.2 Capacitance – Thickness Relationship for Lattice-Integrated dielectric composite

The influence of the 3D pyramidal array on the dielectric behavior of the hybrid joint is presented in Figure 2B, where the equivalent capacitance is plotted against the residual GFRP thickness for both the lattice-integrated specimen and the flat reference. The reference specimen without lattice structures exhibited a standard dielectric response. As the GFRP thickness decreased, its capacitance increased gradually, following the expected inversely proportional relationship ($C \propto 1/d$). The measured values rose from 4.12 pF at 17.20 mm to 10.84 pF at 5.98 mm. Conversely, the specimen featuring the 3D pyramidal array demonstrated a significantly modified electrical response. Even at the maximum thickness of 17.20 mm, the presence of the metallic lattice doubled the initial capacitance compared to the flat reference (7.91 pF vs. 4.12 pF). As the milling process reduced the composite thickness, the capacitance of the lattice-integrated joint exhibited a drastic, non-linear increase. At a residual GFRP thickness of 5.98 mm—just 0.1 mm above the nominal lattice height of 5.88 mm—the equivalent capacitance reached 40.27 pF. This value represents an increase of approximately 270% compared to the flat interface at the same depth. This distinct behavior is directly attributed to the geometry of the interface. The metallic pyramids act as highly localized conductive elements penetrating the dielectric medium. As the top copper electrode approaches the pyramidal tips, the effective distance between the conductive elements is dramatically reduced. This configuration induces a strong local concentration of the electric field around the lattice tips, causing a strongly non-linear increase of capacitance. Finally, at a thickness of 5.88 mm, direct electrical contact with the metallic struts occurred, resulting in a short circuit. These results demonstrate that the embedded metallic lattice markedly modifies the local capacitive field distribution, a key requirement for capacitance-based monitoring of damage-sensitive dielectric changes [15], [16], [17].

3.3 Multi-Channel Capacitive Sensing of Crack Propagation

The results of the Mode I debonding tests performed on the multi-pad hybrid joint are presented through the synchronized analysis of mechanical and capacitive responses in Figure 3, where the correlation between the load-time history, the structural compliance, and the localized capacitance variations (ΔC) across the eight sensing pads is presented. The mechanical load curve exhibits a characteristic saw-tooth profile, where an initial linear increase up to 1.5 kN is followed by a sequence of abrupt load drops. Each of these mechanical events corresponds to a discrete fracture of the interfacial metallic pyramids, characterized by sharp and audible snapping sounds. As these structures fail, the specimen's global compliance increases with temporally aligned step, as highlighted by the gray curve in the lower panel of Figure 3.

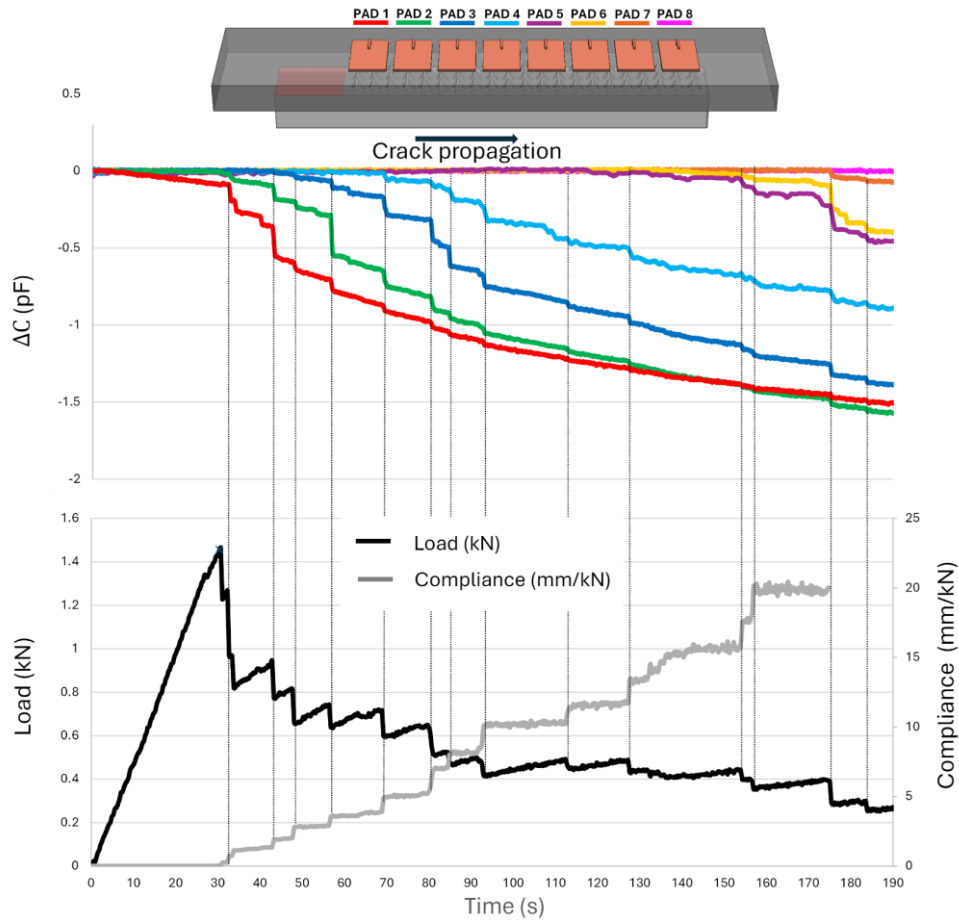


Figure 3. Experimental results for the Mode I DCB test: load-time history (black line) and structural compliance (gray line) plotted against the multi-channel capacitive response (ΔC) for PADS 1–8. Vertical dashed lines highlight the temporal synchronization between mechanical fracture events (load drops/compliance steps) and localized capacitive drops.

The capacitance of each pad generally maintains a stable baseline until the arrival of the crack front. However, as observed in PAD 1, a slight ΔC decrease occurs prior to the first fracture; this initial shift likely reflects the sensing of elasto-plastic deformations within the pyramid struts, causing a slight distancing of the composite from the metallic base before total rupture of the lattice. When the crack front approaches the area below the PAD, a sharp and monotonic drop in capacitance is recorded, temporally synchronized with the load drops and compliance jumps. These events sequentially track crack propagation starting from PAD 1. Although adjacent pads sense nearby failures via electrostatic coupling, the sharpest drop pinpoints the electrode directly above the crack front. Taken together, this information suggests that it may be possible to spatially identify the defect location. Once the crack has moved beyond a sensing area, the ΔC continues to decrease linearly due to increasing distance between the detached adherends. Preliminary tests established that the grounded 3D lattice strongly intensifies the local electric field. Consequently, a fractured pyramid instantly becomes a floating conductor, causing an abrupt charge storage loss that generates a step-wise capacitive drop far exceeding a simple air-gap substitution. Finally, total joint separation occurred just after the initial weak capacitive drop of PAD 7. Supported by visual observations, this suggests that the crack propagated stably up to the region of PADS 5 and 6 and then, after a brief pause, resulted in the sudden and complete detachment of the interface.

Conclusions

This work demonstrates the effectiveness of a multi-channel capacitive SHM system integrated into hybrid metal–GFRP joints with additively manufactured lattice reinforcements. The metallic pyramidal structures act as truly multifunctional features, simultaneously enhancing mechanical performance and amplifying sensitivity to interfacial damage. During Mode I testing, the discrete capacitance drops due to the lattice failure enabled real-time tracking of crack propagation. The proposed approach represents a promising, non-intrusive solution for damage monitoring in high-performance hybrid joints.

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